

Lab 06 - Javascript Home Tasks

1. Write a function `secondLargest(arr)` that returns the second largest unique number from an array. Example: `[10, 5, 8, 10, 3] → 8`.
2. Write a function `countVowels(str)` that returns the number of vowels in a string. Example: `"JavaScript" → 3`.
3. Write a function `reverseWords(sentence)` that reverses the order of words. Example: `"I love JavaScript" → "JavaScript love I"`.
4. Write a function `removeDuplicates(arr)` that returns an array with duplicate values removed. Example: `[1, 2, 2, 3, 1] → [1, 2, 3]`.
5. Write a function `findMissingNumber(arr)` for an array containing numbers from `1` to `n` with one number missing. Example: `[1, 2, 4, 5] → 3`.
6. Write a function `countCharacters(str)` that returns an object containing the frequency of each character. Example: `"aabbcc" → {a: 2, b: 2, c: 1}`.
7. Write a function `longestWord(sentence)` that finds the longest word in a sentence.
8. Write a function `groupEvenOdd(arr)` that returns an object containing separate arrays for even and odd numbers.

```
groupEvenOdd([1, 2, 3, 4, 5, 6]);
```

```
// Output
```

```
{  
  even: [2, 4, 6],  
  odd:  [1, 3, 5]  
}
```

9. Write a function `calculateCartTotal(cart)` that calculates the total price of all products.

```
const cart = [
  { name: "Mouse", price: 1200, quantity: 2 },
  { name: "Keyboard", price: 2500, quantity: 1 }
];
```

10. Create a function `passwordStrength(password)` that returns `"Weak"`, `"Medium"`, or `"Strong"` based on length, uppercase letters, numbers, and special characters.